Lecture 6: Basic Derivative Rules

GOAL: Compute derivatives without relying on the definition.

Summary:

$$\boxed{\begin{array}{ll}
\dfrac{d}{dx}(x^p) = px^{p-1} & \dfrac{d}{dx}(e^x) = e^x \\
\dfrac{d}{dx}(\sin x) = \cos x & \dfrac{d}{dx}(\cos x) = -\sin x \\
\dfrac{d}{dx}(f(x) \pm g(x)) = \dfrac{df}{dx} + \dfrac{dg}{dx} & \dfrac{d}{dx}(k \cdot f(x)) = k\dfrac{df}{dg}
\end{array}}$$